Kwangyoung Lee

HCI researcher, Ph.D Candidate
kwangyoung@kaist.ac.kr
(+82) 10-7347-3676

Research statement

My research focuses on designing interactive systems that help individuals better understand and improve their mental health and well-being, drawing upon Human-Computer Interaction (HCI), Human-Centered Design, and Personal Informatics. I am particularly interested in how subtle and everyday human behaviors or analogies can reveal and externalize invisible personal states, such as emotional experiences and stress, into tangible and reflective data. Building on this, my work supports individuals in making sense of their personal data through structured, goal-oriented approaches and tailored interventions. Recently, I have expanded my research towards understanding how contextual and cultural constraints, such as workplace environments and social media influences, shape personal well-being practices. To address these constraints, I am exploring Collective Personal Informatics, extending individual-level insights to broader ecological systems, and investigating multi-layered interactions between personal data, organizational contexts, and societal influences.

Education

Sep. 2022 - Feb. 2026 (expected)

Korea Advanced Institute of Science and Technology (KAIST)
Ph.D / Industrial Design
Thesis: *Multi-layered Understanding of Personal Informatics: Exploring Contextualized Personal Data Across Intrapersonal, Interpersonal, and Institutional*
Committee: Hwajung Hong (Chair), Taekjin Nam, Uichin Lee, Takyeon Lee, Janghee Cho

Mar. 2016 - Feb. 2018

Ulsan National Institute of Science and Technology (UNIST)
Professional Master of Design-Engineering
Thesis: *Designing Toolkits for Self-Tracking and Self-Intervention to Improve Mental Health*
Advisor: Professor Hwajung Hong

Feb. 2010 - Feb. 2016

Korea Advanced Institute of Science and Technology (KAIST)
Bachelor of Science / Industrial Design

Research Experience

Aug. 2022 - Present

Stress Management by Developing Digital Twins (@ KAIST)

Supported by Institute of Information & communications Technology Planning & Evaluation (IITP)

Role: Project Manager, Research Assistant

- Developing human digital twins technologies for prediction and management of emotional workers' mental health
- Conducting user research to explore how people perceive their emotions in the workplace with a smartphone application though data tracked and how they interact with data to relieve stress

Apr. 2018 - Dec. 2019

Positive Computing through Persuasive Interactions (@ Seoul National University)

Supported by Institute of Information & communications Technology Planning & Evaluation (IITP)

Role: Research Assistant

- Conducting preliminary study on developing tools that can help manage stress by prediction and intervention plan setting
- Designing a calendar-based smartphone application for stress management and conducting user research to understand how people anticipate their daily stress based on the event and how they mediate stress with intervention

Feb. 2017 - Sep. 2017

U-Glass Project (@ UNIST)

Role: Project Manager of design team

- Developed auxiliary tools for an exhibition based on augmented reality (AR) technology
- Managed design team to design content scenario for experiencing museum exhibits with AR glass and conducted user research to explore what will be an engagement factor in an immersive environment

Work Experience

Jan. 2020 - Jul. 2022

Krafton (PUBG Studio), Seoul

Role: UX Designer

- Designed user experience (UX) for lobby screen (outgame) of Battleground Game (Contribution page: Newspage, Notification center, Profile screen, Workshop screen)
- Conducted design research to induce user retention

Jan. 2017 - Feb. 2017

ITNJ (Start-up), Ulsan

Role: UX Research Assistant / Product Designer

- Developed online educational content and designed portable tablet cradle
- Discovered new businesses based on the experiences of people using tablet PCs and designed the cradle to install the tablet

Jun. 2014 - Dec. 2015

Coin-Cloud (Start-up), Daejeon
Role: CEO / Designer

- Developed coin accumulation system using smartphones to reduce coin issuance cost
- Participated in start-up competition hosted by KBD Industrial Bank and attracted investment from companies

Publication

DIS 2025 (Conditionally Accepted)

Thinking Outside the Data Box: Investigating the Potential of Data Manipulation for Self-Reflection on Personal Data
Yeohyun Jung, **Kwangyoung Lee**, Hwajung Hong
DIS '17: Proceedings of the 2017 Conference on Designing Interactive Systems

CHI 2025 (Accepted)

Peerspective: A Study on Reciprocal Tracking for Self-awareness and Relational Insight
Kwangyoung Lee, Yeohyun Jung, Gyuwon Jung, Xi Lu, Hwajung Hong
CHI '25: Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems

BigComp 2023 (Workshop)

Designing a Personalized Stress Management System for Call Center Workers
Kwangyoung Lee, Hyunseung Lim, Sooyeon Ahn, Taewan Kim, Hwajung Hong
2023 IEEE International Conference on Big Data and Smart Computing (BigComp)

CHI 2020

Toward future-centric personal informatics: Expecting stressful events and preparing personalized interventions in stress management
Kwangyoung Lee, Hyewon Cho, Kobiljon Toshnazarov, Nematjon Narziev, So Young Rhim, Kyungsik Han, YoungTae Noh, Hwajung Hong
CHI '20: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems

DESIGN 2018

Aesthetic product interaction: The necessity of consistency between function and emotion
K. Lee, J. A. Self, H. Hong
DS 92: Proceedings of the DESIGN 2018 15th International Design Conference

CHI 2018

MindNavigator: Exploring the stress and self-interventions for mental wellness
Kwangyoung Lee, Hwajung Hong
CHI '18: Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems

DIS 2017

Designing for self-tracking of emotion and experience with tangible modality
Kwangyoung Lee, Hwajung Hong
DIS '17: Proceedings of the 2017 Conference on Designing Interactive Systems

Teaching Experience

Teaching Assistant

Fall, 2016

Design Knowledge and Skills (Intro to Information Visualization)
Course: IID231, +20 students, UNIST

- An undergraduate course for teaching basic knowledge, tools, and practical skills for visualizing information with data.
- Provided feedback in the weekly one-on-one instruction session, guiding individual project development.

Spring, 2023

Needfinding Practice
Course: KEI430, +20 students, KAIST

- An undergraduate/graduate course for teaching design thinking based on observation of user behaviors.
- Provided feedback on the weekly group project development.

Award

2018

Outstanding Presentation Award
ACM SIGCHI KOREA LOCAL CHAPTER

Academic service

Peer reviewing

ACM Human Factors in Computing Systems (CHI) 2024, 2025
ACM Designing Interactive Systems (DIS PWIP). 2025
Archive of Design Research (AODR). 2023, 2024, 2025